

Research article

Combining deep learning and machine learning techniques to track air pollution in relation to vegetation cover utilizing remotely sensed data

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ABSTRACT

The rapid urban expansion in Dhaka, the capital of Bangladesh, has escalated air pollution levels and led to a significant decrease in green spaces. This study employed machine learning (ML) and deep learning (DL) techniques to examine the relationship between rising concentrations of particulate matter (PM_{2.5} and PM₁₀) and decreasing urban green spaces from 1990 to 2022. The ML algorithms, specifically XGB, SVM, and RF, effectively predicted high air pollution areas, while DL models Unet, Unet++, MAnet, and Linknet accurately forecasted vegetation cover trends. The findings confirm a strong negative correlation between increased air pollution and vegetation. The decline in green spaces is not only a local concern but also has broader regional implications due to the transboundary nature of air pollution. The results highlight the critical need for pollution management strategies and urban planning that prioritize green infrastructure. The study also emphasizes the value of using ML and DL techniques for accurate, data-driven environmental assessments and predictions. Future studies could incorporate high-resolution images and integrate socioeconomic data to achieve a more comprehensive perspective on the urban environmental challenges faced by rapidly developing cities like Dhaka. The use of an integrated ML and DL strategy as highlighted in this research appears to be a practical and economical method for tracking vegetation degradation and change, and in establishing the causal links to air pollution.

1. Introduction

At present over half the world's population lives in urban areas, and this number is projected to increase to close to 70% by 2050 (Han et al., 2022; Lu et al., 2024). On a related note, air pollution, and particulate matter (PM) in particular, is now considered to be the single largest environmental health risk to humans, causing over 4 million premature deaths every year (Diener and Mudu, 2021; Han et al., 2022; Rai, 2016). Environmental changes due to PM in urban areas, exert a significant impact on urban vegetation, in terms of its health, morphology and physiological status (Rai, 2016; Su et al., 2022). Specifically, vegetation has been shown to reduce ambient PM concentrations due to adsorption, although capacity varies with vegetation type and morphology (Li et al., 2023; Su et al., 2022; J. Zhang et al., 2024). This allows researchers to

evaluate the characteristics of urban vegetation as an indicator of PM pollution impacts across urban landscapes and has been found effective for such endeavors (Rai, 2016; J. Zhang et al., 2024). A better understanding of the urban vegetation landscape and the direct impact of green spaces' reductive effects on such PM based air pollution, can thus lead urban planners and managers to more effectively manage these complex systems for the benefit of the human populations living within them (Diener and Mudu, 2021; Li et al., 2023; J. Zhang et al., 2024).

The interplay between vegetation health and air quality forms a critical aspect of environmental science, with one inevitably influencing the other. Vegetation, acting as Earth's natural lungs, plays a pivotal role in modulating air quality, yet remains susceptible to the adverse effects of air pollution. A decrease in vegetation health and coverage can increase atmospheric pollutants, forming a cycle that harms both

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ecosystems and human health. Air pollution, characterized by harmful substances in the air, significantly affects human health, the environment, and living organisms (Kampa and Castanas, 2008). The World Health Organization (2022) estimates that air pollution causes approximately 7 million premature deaths each year. Air pollution arises from both natural sources like dust and wildfires and human activities such as industry, transport, and energy production. The range of pollutants in our air is broad, including substances such as particulate matter, nitrogen oxides, ozone, sulfur dioxide, and carbon monoxide. These pollutants, aside from being direct health hazards, bear significant detrimental implications for vegetation (T.-M. Chen et al., 2007; Cincinelli and Martellini, 2017; Z. Wang et al., 2019). The United States Environmental Protection Agency underscores the hazards of particulate matter (PM), particularly PM₁₀ and PM_{2.5}, for their ability to deeply infiltrate respiratory systems and infiltrate the bloodstream. Additionally, these fine particles, contingent on their chemical make-up, can pose extensive threats to ecosystems, modifying soil nutrient content, contaminating water bodies, harming agricultural output, and inducing acidic precipitation (US EPA, 2016).

Vegetation, as an indicator of environmental health, has a complex interaction with air pollutants. Adverse interactions can manifest in various manners: pollutants settling on plants can obstruct photosynthesis; shifts in soil chemistry due to pollutants can inhibit nutrient uptake; alterations in the atmospheric composition can modify weather patterns affecting plant growth. Numerous studies corroborate these findings, showcasing how pollutants like ozone (O₃), nitrogen dioxide (NO₂), and sulfur dioxide (SO₂) stymie growth and compromise plant vitality (Allen, 1990; D'amato et al., 2001). Ozone stands out as a particularly harmful air pollutant. Elevated ozone levels inflict damage on plant cellular structures, inhibiting photosynthesis, and stunting growth (Morgan et al., 2003). Such interference can culminate in reduced agricultural output and heightened vulnerability of plants to external threats, including pests, diseases, and environmental stressors (Mudd, 2012). Additional pollutants, including SO₂ and nitrogen oxides, can exacerbate soil acidification, undermining plants' nutrient absorption capabilities (Varshney et al., 1979; Shammas et al., 2020).

Particulate matter (PM), comprising solid particles and liquid droplets in the air, includes the fine particles of PM_{2.5}, with diameters of 2.5 µm or less, and the coarser PM₁₀ particles, with diameters up to 10 µm. PM_{2.5} particles arise primarily from combustion activities, such as vehicle emissions, industrial processes, and burning of biomass, while PM₁₀ can also include larger particles like dust, pollen, and mold, often suspended in the air due to construction, agriculture, and natural disturbances (Kampa and Castanas, 2008). These particles are not only a concern for human health, penetrating deep into the lungs and even entering the bloodstream to cause respiratory and cardiovascular issues, but they also have a detrimental impact on vegetation (Shahriar et al., 2020). The deposition of PM on leaves can obstruct sunlight, impairing the process of photosynthesis. Furthermore, the absorption of these particles by plant stomata can lead to toxic effects, reducing growth rates, damaging leaf structures, and ultimately affecting ecosystem health and crop yields (Su et al., 2022; Han et al., 2022; Diener and Mudu, 2021; Mudd, 2012). Monitoring the levels of PM₁₀ and PM_{2.5} is thus essential for safeguarding both public health and the environment, guiding policies to mitigate air pollution's wide-reaching effects.

In Bangladesh, air pollution is a growing environmental concern. Factors such as industrial activities, vehicle emissions, and household energy use significantly contribute to the region's declining air quality (Hasnat et al., 2018; Khallaf, 2011). This increase in pollution impacts food security and affects agricultural productivity (Ghose, 2014; Hasnat et al., 2018). Recent research efforts have focused on understanding the effects of air pollution on the country's vegetation and crops (Khuda, 2020). A notable study by Shahriar et al. (2020) adopted a range of computational methodologies, including linear and Gaussian support vector machines, Gaussian process regression, random forest regression, and artificial neural networks, to assess the atmospheric concentrations

of PM_{2.5} and PM₁₀. Their findings highlighted seasonal variations with pollutant concentrations peaking during winter and plummeting in the monsoon season. Echoing this, Begum et al. (2011) identified exacerbated air quality concerns during dry winter months in Dhaka, attributing it to elevated particulate matter concentrations. Finally, a study by Ahmed et al. (2022) underscored the ramifications of household air pollution on women's health, emphasizing the choice of cooking fuels and cooking locations using MANOVA.

Broadening the scope to encompass global studies provides insights into the spatial and temporal effects of air pollution on vegetation. For instance, Liu et al. (2021b) undertook an assessment in California to understand the spatiotemporal repercussions of the COVID-19 pandemic on air pollution. By comparing air pollutant data from 2020 against an average from 2015 to 2019, the study showcased the influence of lockdown and reopening strategies on air quality. Notably, carbon monoxide levels experienced a more pronounced decline compared to NO₂ and PM_{2.5} during the pandemic period. Deshmukh et al. (2019) embarked on a study to ascertain the impact of roadside vegetation on proximate air quality. Their observations along highways with diverse vegetation characteristics revealed that dense, tall, and continuous vegetation stands significantly curtailed downwind pollutants, emphasizing the importance of strategic vegetation planning. Gong et al. (2023), employing a Bayesian meta-analysis approach, reinforced the notion that urban vegetation serves as a viable mechanism to diminish concentrations of pollutants such as PM, NO, SO₂, and O₃. While Bangladesh serves as the primary focus of this research, understanding global trends and methodologies offers a comprehensive perspective on the intricate nexus between air pollution and vegetation health.

In the realm of predicting vegetation dynamics and its relationship with environmental variables, advanced computational techniques have gained substantial prominence. Machine learning methodologies, distinguished by their adaptability and precision, have been at the forefront of these endeavors. Among them, techniques such as the Least Absolute Shrinkage and Selection Operator (LASSO), Random Forest, and Long Short-Term Memory Networks (LSTM) have been employed to forecast crop yields with notable success. Deep learning models, a subset of machine learning, further enhance predictive capabilities with their intricate architectures tailored for complex data relationships. Key models in this domain, like the Fully Connected Long Short-Term Memory (FC-LSTM), Convolutional Long Short-Term Memory (ConvLSTM), and Vanilla Recurrent Neural Network (RNN), have been instrumental in deciphering the intricate correlations between vegetation and environmental variables, including factors like precipitation and fire occurrences. Extending beyond vegetation, several studies have sought to understand air pollution dynamics through machine learning. Deep learning models, particularly LSTM and Convolution Neural Networks (CNN), have been adopted to predict specific pollutant concentrations, such as PM_{2.5}, and more broadly, air quality classifications. Augmenting this, some studies have innovatively merged deep learning techniques with chemical indicators to deliver accurate estimations of air quality coefficients (Bao et al. (2022)).

The rapid expansion of urban and industrial areas in Bangladesh has significantly contributed to the deterioration of vegetation due to escalating air pollution levels, as noted in studies by (Hagemann et al., 2018; Hossain and Huggins, 2021). This study seeks to explore the connection between air pollution and vegetation coverage, employing machine learning (ML) and deep learning (DL) approaches. To bridge the gaps identified in previous research, we utilize advanced DL segmentation models, including Unet, MAAnet, and DeepLabV3+, to identify areas of dense vegetation in Dhaka. Simultaneously, ML methods such as support vector machines, k-nearest neighbors, and k-means are used to pinpoint regions with high levels of air pollution. Spanning from 1990 to 2022, our approach integrates geospatial techniques with ML and DL models to examine the relationship between areas characterized by lush vegetation and those affected by severe air pollution. This analysis allows us to evaluate the effectiveness of both ML and DL models in

predicting these environmental features and to explore the spatial correlation between vegetation density and air pollution across Dhaka. Our research questions are: (1) what is the ability of ML models to forecast high pollution areas in Dhaka and can we identify the most accurate model; (2) what is the precision of the DL segmentation models in mapping dense vegetation cover and how can we determine the most reliable model; and (3) what is the spatial interaction between high vegetation and pollution areas as predicted by these models. This study is pivotal in addressing the adverse impacts of air pollution on vegetation, crucial for sustainable urban planning and environmental preservation in Bangladesh. By leveraging sophisticated computational techniques, our findings aim to influence policymaking and set a foundation for future environmental protection and restoration initiatives.

2. Methodology

2.1. Study area

The focal point of our research is Dhaka city, the capital of Bangladesh (Fig. 1). Notably, Dhaka’s air quality has deteriorated significantly over the years, earning it a ranking as the third worst on the global Air Quality Index (AQI). This classification highlights the city’s struggle with extreme air pollution, situating it among the most polluted urban centers globally. Based on the US EPA Standards (EPA, 2024) Dhaka experiences PM ranges identified as ‘High’ for the majority of the winter season. A significant concern for Dhaka, akin to numerous other urban areas in south and southeast Asia, is the elevated concentration of PM in its atmosphere (Rana et al., 2016). Historical analysis reveals that

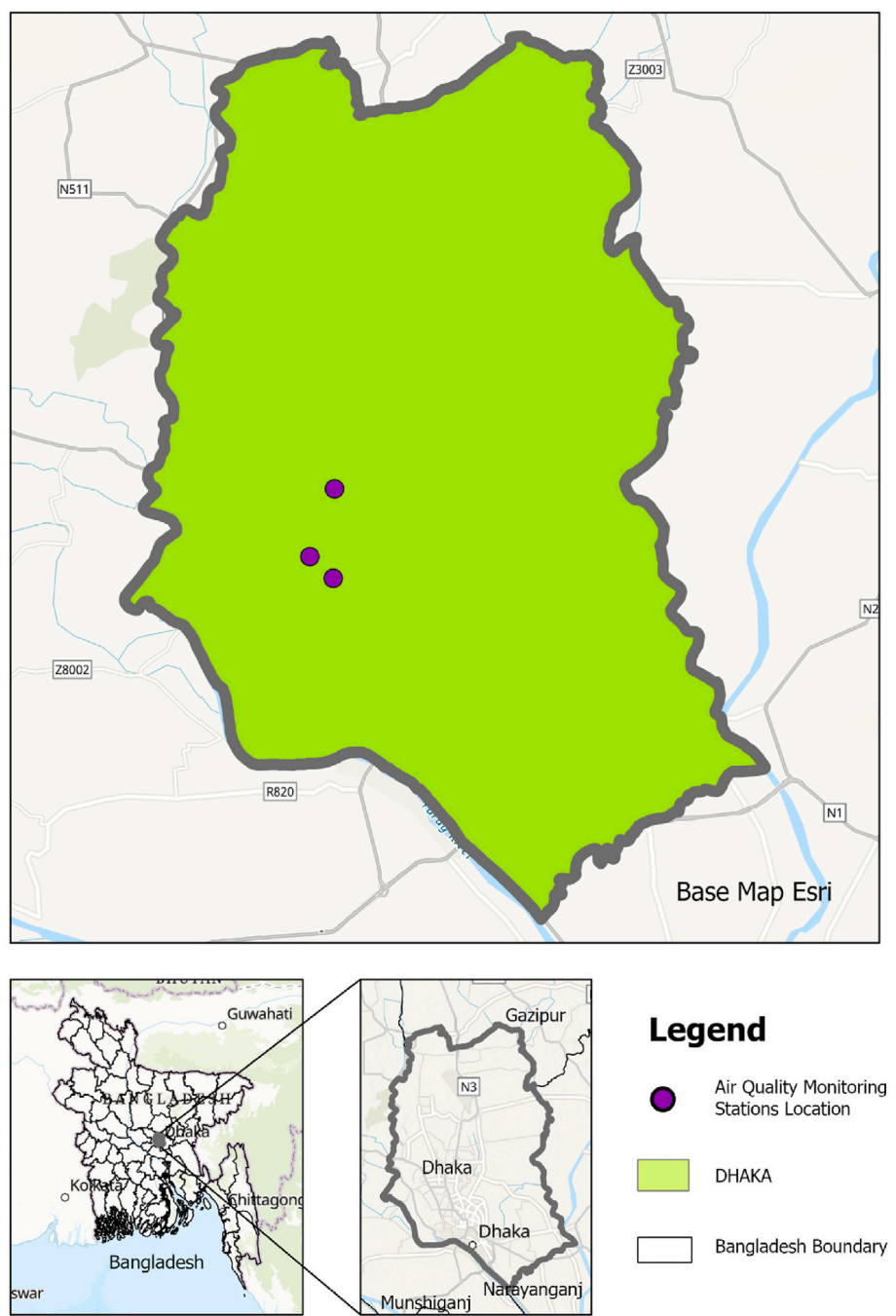


Fig. 1. Geospatial map of Dhaka within Bangladesh, highlighting its central position and the locations of the air quality monitoring stations.

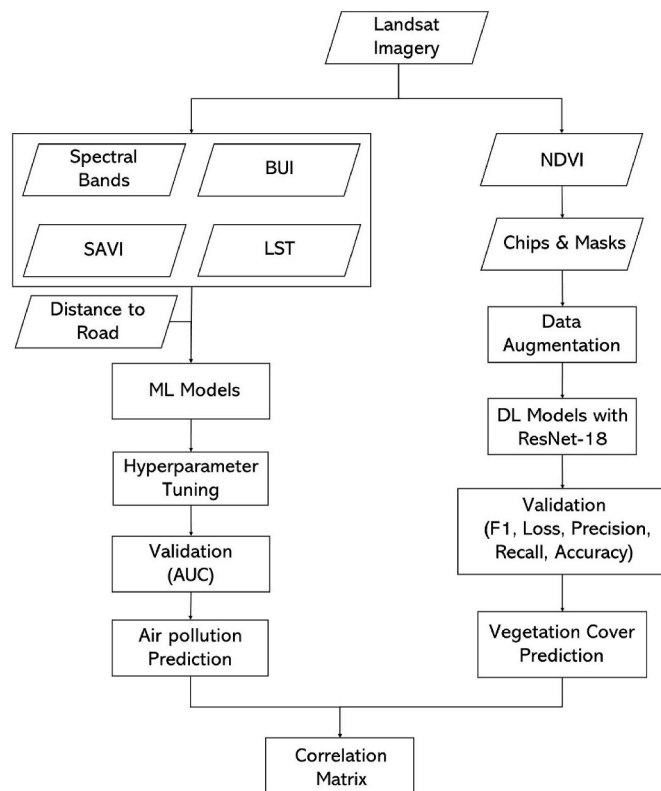


Fig. 2. Overall Methodological Framework indicating the steps and layers created as part of the analysis process.

a vast expanse, approximately 80% of Greater Dhaka, was predominantly rural. However, rapid urbanization and environmental disruptions have not only altered the city's ecosystem but have also influenced its climatic patterns (M Kamal Pasha et al., 2022). Urban planning in Dhaka has often overlooked the importance of green spaces. The city's road infrastructure largely lacks designated zones for tree planting. As a result, the tree coverage varies considerably across different parts of the city. For instance, bustling commercial zones are predominantly devoid of trees (Nigar Neema and Jahan, 2014). To facilitate regular air quality assessments, Dhaka is equipped with three air quality monitoring stations (Fig. 1). These stations play a pivotal role in providing real-time data on the city's air quality metrics.

2.2. Predicting areas of high air pollution using ML models

2.2.1. Air quality & satellite imagery data collection

The core of our investigation on air pollution in Dhaka revolves around a concentrated study period spanning three months: December, January, and February. This specific timeframe was chosen based on multiple studies that consistently indicate a surge in air pollution levels during these winter months (Bodor et al., 2020; Lin et al., 2009; Phung-Ngoc et al., 2023). Winter in Dhaka is characterized by atmospheric conditions that trap pollutants closer to the ground, leading to reduced air quality and heightened pollution levels. Complementing our air pollution data, we sourced satellite imagery from the Landsat program. Landsat satellites, operated by NASA and the USGS, have been capturing Earth imagery for decades, providing an invaluable repository for environmental research. For our study, we procured Landsat 8 imagery corresponding to the selected winter months, ensuring that our satellite data aligns temporally with our air quality measurements. This imagery offers a comprehensive view of land cover changes, and the vegetation dynamics, to allow us to characterize the vegetation cover for the same time of year as we have measured air pollution values.

This study utilizes data from three air quality monitoring stations in Dhaka, Bangladesh, to analyze air pollution trends from 2015 to 2018. These stations are CAMS-1 located in Sangsad Bhaban, Sher-e-Bangla Nagar; CAMS-2 in Firmgate; and CAMS-3 in Darus-Salam (Fig. 1). We collected monthly data on two key air pollution metrics: the maximum concentrations of PM_{2.5} and PM₁₀. This data was obtained from the official website of the Department of Environment, Bangladesh (<https://doe.portal.gov.bd/>). To ensure a comprehensive analysis, we included data from all available months within the study period, instead of limiting our analysis to a three-month dataset. It is important to note that our data corroborate the findings of other researchers, highlighting the winter maximums of air pollution for the PM (Table 1). A critical aspect of our methodology involves calculating the ratio of PM_{2.5} to PM₁₀ for each month. This ratio is instrumental in discerning the primary sources of particulate matter by indicating the relative presence of fine versus coarse particles. According to Spandana et al. (2021), a higher PM_{2.5}/PM₁₀ ratio signifies a greater potential for severe air pollution due to the increased harmfulness of finer particles. Zhao et al. (2019) emphasize the importance of reducing the proportion of PM_{2.5} within PM₁₀ to mitigate pollution effectively. Given Dhaka's existing high pollution levels, we applied this ratio as a key factor in our modeling approach. Following the calculation of monthly PM_{2.5}/PM₁₀ ratios for each station, we computed the average ratio for each station over the four-year period, given air pollution impacts on vegetation are cumulative over time (Table 1).

The modeling methodology utilized three distinct threshold values to categorize the air pollution zones into low, medium, and high classifications (see Table 1). The categorization of each station was established based on the data presented in Table 1 and linked to the literature. When comparing across the three sites for the PM_{2.5}/PM₁₀ ratios, we can determine that CAMS3 has the lowest level of 0.51, CAMS1 has a medium level with 0.61, and CAMS2 is the highest with 0.66 (Table 1). These station levels, as calculated from the station data for 2015–2018, serve as a standard reference for the machine learning analysis performed.

The Landsat satellite imagery was collected specifically for the winter season due to the documented high levels of pollution during this time period, as mentioned in the existing literature (Hien et al., 2011) and again substantiated by this research (Table 1). The time period from 2015 to 2018 was selected due to the availability of sufficient data on PM_{2.5} and PM₁₀ for the purposes of this study on Dhaka City. Landsat 8 image bands 2–6, as well as derived Spectral Vegetation Index (SAVI) and Built-Up Index (BUI) data, was utilized (Fig. 3). The BUI helps in identifying and mapping urban areas. It leverages the spectral reflectance properties of typical urban materials such as asphalt, concrete, and buildings. A higher BUI value typically indicates more densely built-up areas, which can be sources of pollution due to vehicular traffic, industries, and more. The SAVI is a vegetation index designed to minimize the influence of soil brightness when vegetation cover is low. By providing a measure that is sensitive to the density of green vegetation, SAVI is useful for monitoring vegetation health, potential stress or degradation, and its interactions with atmospheric pollutants. The land surface temperature was also calculated as this exhibits a significant correlation with air quality (Parida et al., 2021) and was derived according to the methods employed by (Das and Das, 2020). The data pertaining to street networks was obtained from OpenStreetMap (<https://www.openstreetmap.org/#map=5/38.007/-95.844>) for all primary routes and distance to road measures created. Distance to Road was calculated to measure the proximity of a point to the nearest road as areas closer to major roads can be more exposed to vehicular emissions, a significant contributor to air pollution. Proximity to roads can have implications for vegetation health, given the potential for higher pollutant concentrations. All data were created for the Dec–Feb season, across the 2015–2018 time period, to match the air pollution data collection periods. Subsequently, a composite raster was generated by computing the average of the three seasons and this single raster was

Table 1

Monthly averaged PM_{2.5}, PM₁₀ and the ratio of PM_{2.5}/PM₁₀ data averaged across 2015–2018 for each of the three air pollution collection stations in Dhaka: CAMS-1 located in Sangsad Bhaban, Sher-e-Bangla Nagar; CAMS-2 in Firmgate; and CAMS-3 in Darus-Salam. For the three stations the overall PM_{2.5}/PM₁₀ ratio averaged across all months, and its level is also indicated.

Month	CAMS1 PM _{2.5} MAX	CAMS1 PM ₁₀ MAX	CAMS1 PM _{2.5} / PM ₁₀	CAMS2 PM _{2.5} MAX	CAMS2 PM ₁₀ MAX	CAMS2 PM _{2.5} / PM ₁₀	CAMS3 PM _{2.5} MAX	CAMS3 PM ₁₀ MAX	CAMS3 PM _{2.5} / PM ₁₀
JAN	167	241	0.69	176	269	0.65	212	335	0.63
FEB	130	201	0.65	153	420	0.36	160	321	0.50
MAR	84.8	165	0.51	185	296	0.63	123	270	0.46
APR	46.1	88.9	0.52	84.49	145.4	0.58	74.89	140.02	0.53
MAY	46.2	77.3	0.60	74	123	0.60	53.3	100	0.53
JUNE	38.5	N/A	N/A	47.8	71.8	0.67	37.7	77.7	0.49
JULY	25.1	45.3	0.55	41.2	64.1	0.64	32.6	91.4	0.36
AUG	28.67	53.06	0.54	41.3	63.36	0.65	33.16	72.66	0.46
SEPT	36.5	N/A	N/A	55.9	77.4	0.72	52.2	113	0.46
OCT	68.4	97.3	0.70	67.6	95.8	0.71	63.1	131	0.48
NOV	87.6	136	0.64	112	262	0.43	121	197	0.61
DEC	163	231	0.71	235	175	1.34	190	291	0.65
AVG.			0.61 MEDIUM			0.66 HIGH			0.51 LOW

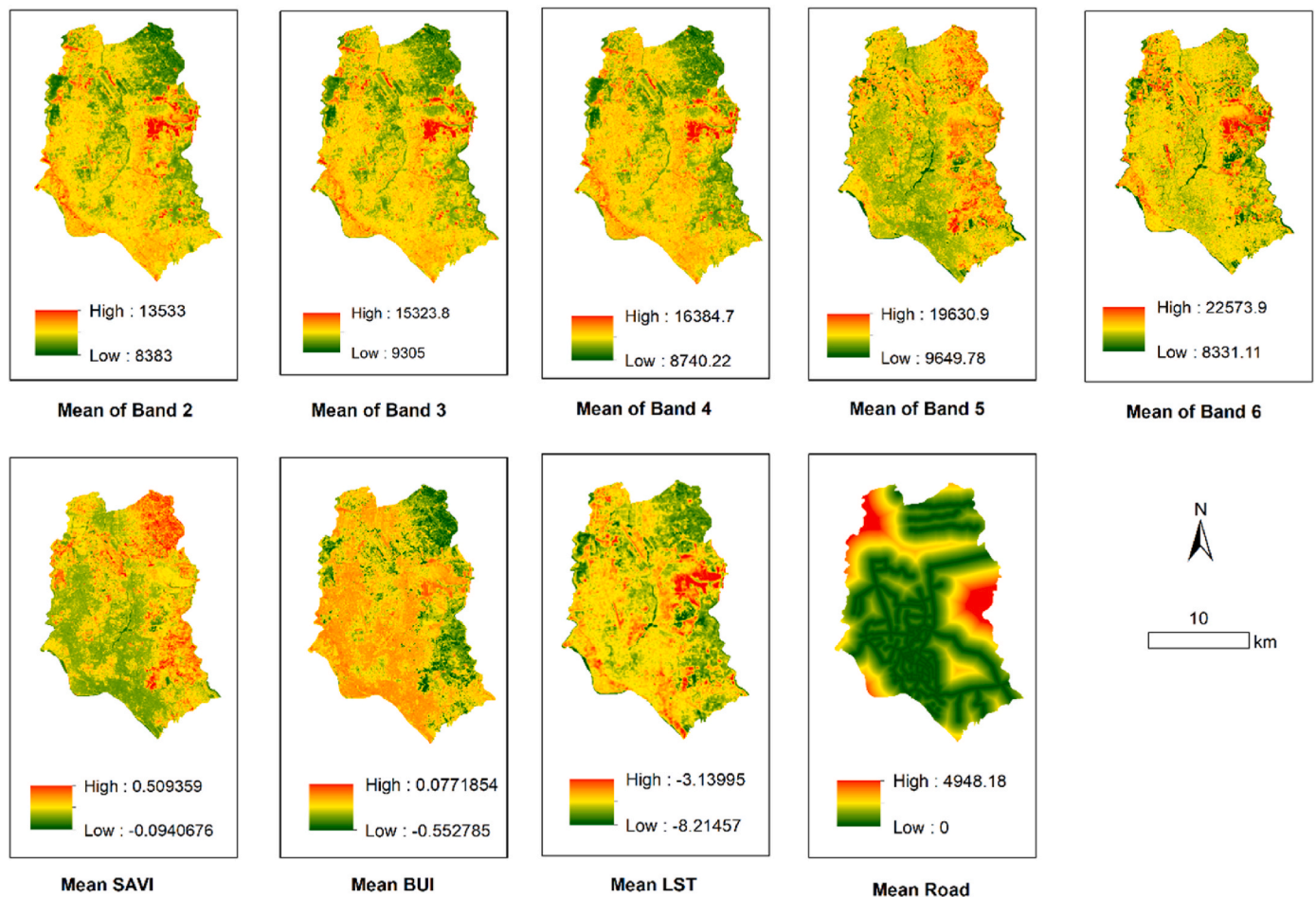


Fig. 3. Variables used in Machine Learning models to predict areas of high area pollution shown as their mean surface values, as calculated from the Landsat 8 satellite images, and roads layer for distance to roads surface.

employed for each variable during the modeling procedure (Fig. 3). These variables are all used to predict the locations of high air pollution in Dhaka.

Ensuring the precision and dependability of satellite imagery and environmental data is crucial for efficient environmental surveillance and modeling, especially for activities like identifying areas with heightened concentrations of PM_{2.5} and PM₁₀ particles. Satellite data obtained from initiatives such as Landsat offer crucial and uninterrupted

observations over extended periods, covering a wide range of geographical, temporal, and spectral resolutions. Nevertheless, these datasets present difficulties such as cloud cover, spectrum mixing, and the requirement for meticulous preprocessing techniques like radiometric calibration and atmospheric correction to guarantee the best representation of the Earth's surface. Environmental data pertaining to air quality is reliably monitored by ground-based monitoring stations, which are overseen by agencies such as the US EPA. However, these

metrics may be limited in their ability to accurately represent different locations and their frequency over time. These stations are commonly situated in metropolitan areas, which can create prejudice based on geography, and the frequency of data collection may not precisely catch brief occurrences of pollution. To reduce these possible biases, a variety of solutions are implemented. These strategies involve the fusion of data, which involves integrating several satellite datasets, and using advanced ML methods for preprocessing. Furthermore, routine calibration and repair of ground equipment are carried out, and monitoring networks are developed to span diverse locations. In addition, utilizing established methodologies for gathering and analyzing data reduces the probability of errors caused by human and operational factors. The adoption of these comprehensive approaches enhances the precision and reliability of environmental monitoring and modeling outcomes by improving both satellite and ground-based environmental data (Fig. 3).

2.3. Machine learning models

Past research initiatives have harnessed the capabilities of ML algorithms to predict and analyze air pollution patterns with considerable success (Balogun et al., 2021a; Doreswamy et al., 2020; Shogrkhodaei et al., 2021; Tella and Balogun, 2022). Compared to traditional models, ML models often offer superior performance while retaining a level of simplicity and efficiency (Mosavi et al., 2018). By leveraging these algorithms, we aim to generate a comprehensive prediction of high air pollution zones in Dhaka. This study utilizes a range of machine learning algorithms to predict air quality zones, each rooted in distinct computational principles. The Support Vector Machine (SVM), known for its effectiveness in classification and regression through optimal hyperplanes and kernel functions, is implemented in R software (Balogun et al., 2021b; Cervantes et al., 2020; Cortes and Vapnik, 1995; Vapnik, 1999). Extreme Gradient Boosting (XGBoost) leverages parallel computing to enhance decision tree performance, focusing on minimizing previous errors and offering tools for feature importance (Chang et al., 2018; Ibrahim Ahmed Osman et al., 2021). The K-Nearest Neighbors (KNN) algorithm classifies data points based on the classification of their nearest neighbors, requiring careful selection of 'k' and a distance metric (Gallego et al., 2022; Gou et al., 2019; Y. Wang et al., 2022). Naive Bayes (NB) utilizes Bayes' theorem to provide probabilistic predictions, assuming conditional independence among features (Kumar and Pande, 2023; Sethi and Mittal, 2022). Decision Trees (DT) recursively split data to build a tree-like model, prioritizing features for informative splits (Otukey and Blaschke, 2010; Pal and Mather, 2003; Phiri et al., 2020). Random Forest (RF) combines multiple decision trees to counter overfitting and improve prediction accuracy and feature importance ranking (Balogun et al., 2021b; Breiman, 2001; Peters et al., 2007). Lastly, Artificial Neural Networks (ANNs) mimic biological neural networks to identify complex patterns, suitable for datasets with non-linear associations and varied distributions (Azid et al., 2014; Cabaneros et al., 2019; Lek and Guégan, 1999; Maleki et al., 2019; G. Zhang et al., 1998).

2.4. Deep learning model predictions of vegetation cover across the Dhaka region

2.4.1. Landsat satellite input image data

Vegetation cover was monitored using DL segmentation models to predict areas of high vegetation cover across the Dhaka study area. For the training of the deep learning model, satellite images were collected every three years from 1990 to 2020. For Landsat 5 data images used were from the following dates: 1990-01-07, 1993-11-15, 1996-12-25, 1999-12-18, 2002-12-18, 2005-01-19, 2008-12-10, and 2011-01-01. For the Landsat 8 data the images used were from the following dates: 2014-01-25, 2017-12-03, and 2020-12-27. The selection of bands 5, 4, and 3 for Landsat 5 and bands 6, 5, and 4 for Landsat 8 aimed to capture data from the visible and near-infrared spectrum, crucial for assessing

vegetation. Despite the different band numbers, these selections were based on their sensitivity to vegetation coverage, acknowledging that spectral responses might vary due to the distinct sensor technologies and designs of the Landsat missions.

The analysis process involved standardizing the satellite images into RGB composites and converting them to an unsigned 8-bit integer format for uniformity. An important procedure was the creation of a vegetation mask by the categorizing of NDVI values. Vegetation was indicated by NDVI values larger than 3000, as we employed 16-bit integer NDVI images with values ranging from -10000 to $+10000$. The focus was on the high vegetation cover class for extracting training data for the DL models. To facilitate detailed and efficient analysis, the study area's imagery was segmented into smaller sections or "chips" of 32×32 pixels, enhancing the dataset's granularity and the manageability of computational processes. These chips, representing areas of the highest vegetation cover as determined by the NDVI mask, were then used to train the DL models, focusing solely on the extracted bands pertinent to vegetation (Fig. 4). This methodological approach ensured a focused and detailed analysis of vegetation cover over time, leveraging advanced DL techniques for segmentation and analysis.

2.4.2. Configuration of the deep learning model

For the deep learning model, we utilized a ResNet-18 architecture, optimized for our dataset's limited size. This model, pre-loaded with ImageNet-trained weights, leverages a sigmoid activation function to fit binary classification by normalizing output values within a 0 to 1 range. The input images, prepared as $32 \times 32 \times 3$ chips to represent RGB values, undergo processing with a stride of 8 and a central crop of 12 pixels to enhance focus on relevant image areas. The architecture channels these images through four residual blocks, reducing spatial dimensions while increasing depth, from Layer 1 to Layer 4 (Fig. 5). This journey through the blocks culminates in an average pooling layer that simplifies the feature maps into a format suitable for a sigmoid activated fully connected layer, resulting in a comprehensive $1 \times 1 \times 512$ output. This methodical design ensures the model proficiently extracts hierarchical features, essential for identifying complex patterns within the data, facilitating accurate classification tasks.

2.4.3. Deep learning models employed

DL employs intricate architectures that automatically learn patterns directly from data, especially large-scale datasets. In the realm of image analysis, DL models have excelled due to their capacity to recognize complex patterns, hierarchical features, and contextual information within images. In our study, we harness the power of DL for geospatial segmentation tasks. Specifically, our objective revolves around predicting the presence of vegetation cover. This study utilizes a range of DL models for image segmentation tasks, each with unique architectures and strengths. The UNet model is known for its precision in pixel-wise classification due to its effective integration of low and high-level features (Jiao et al., 2020; Wang et al., 2022). Enhancing UNet, the UNet++ model introduces dense skip connections for multi-scale feature extraction, with a structure that includes a single encoder and multiple decoders (Peng et al., 2019; Bao et al., 2021; Liu et al., 2021a). The Multi-scale Attention Network (MANet) is designed for semantic segmentation of high-resolution imagery, utilizing a dual-layer channel attention mechanism for feature refinement (Chen et al., 2022). LinkNet evolves from UNet with a U-shaped structure that minimizes spatial information loss by sharing learnings between the encoder and decoder (Guo et al., 2021; Wu et al., 2021). The Feature Pyramid Network (FPN) constructs a feature pyramid with a top-down mechanism for rich semantic predictions (Handani et al., 2021), while the Pyramid Scene Parsing Network (PSPNet) combines global and local contexts for per-pixel classification (Handani et al., 2021; Fang et al., 2021). DeepLabV3 and DeepLabV3+ leverage atrous convolution for feature extraction at multiple resolutions, with the latter adding an encoder-decoder structure for enhanced boundary delineation and

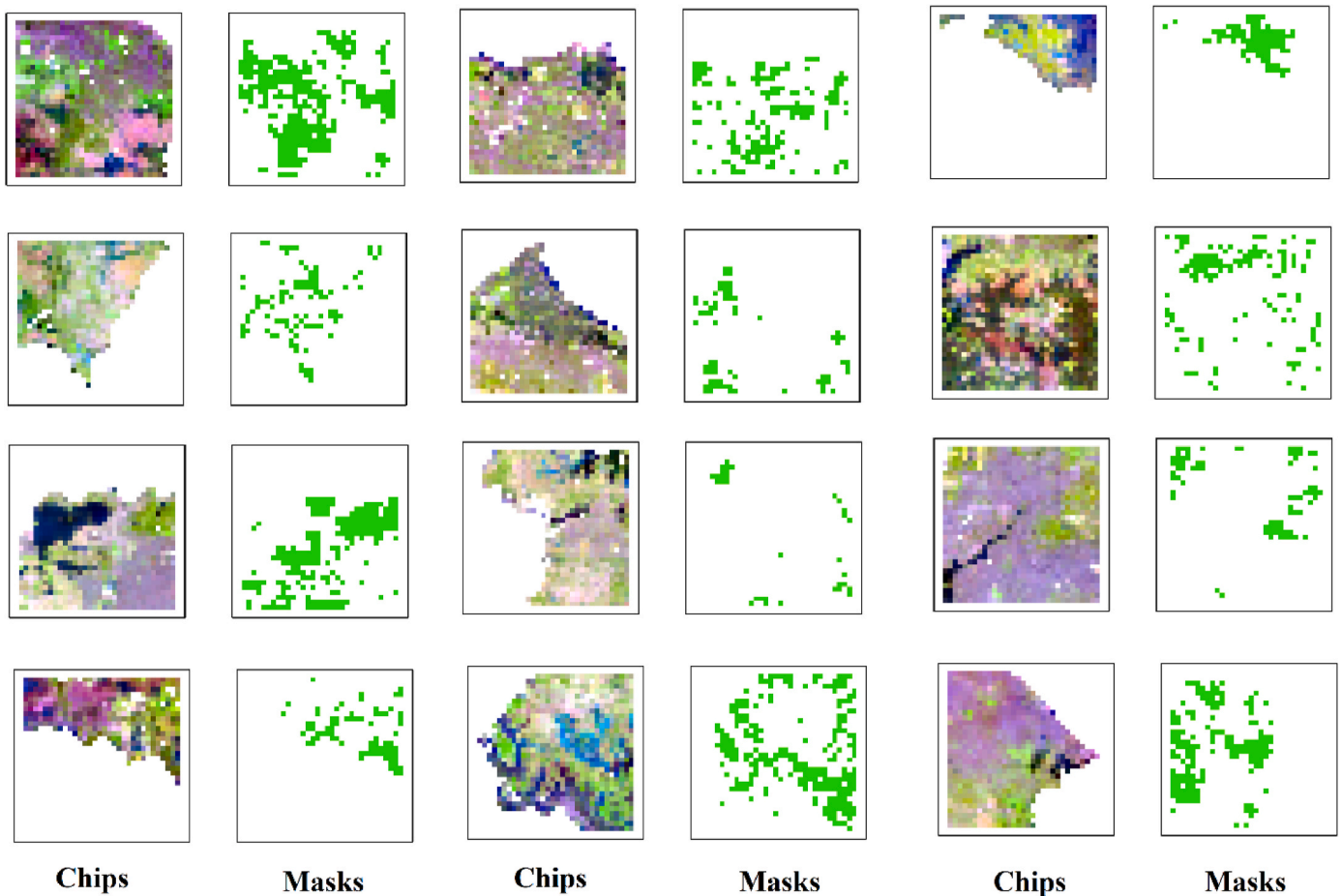


Fig. 4. Extracted image chips and corresponding NDVI vegetation masks utilized in training deep learning models. Only data shown in the mask is extracted from the chip layer.

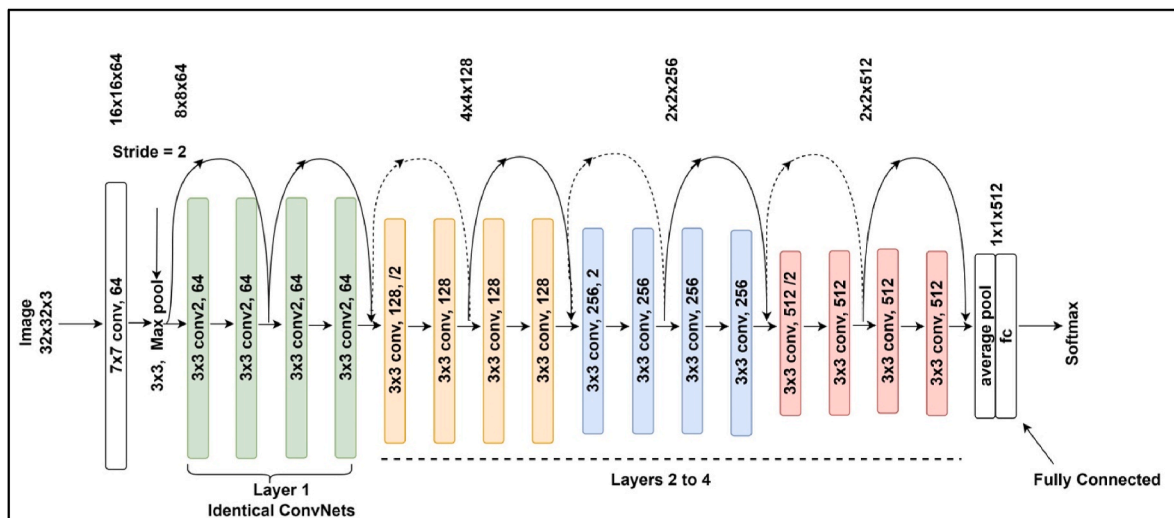


Fig. 5. Schematic diagram of ResNet-18 neural network architecture.

multi-scale context information (Handani et al., 2021; Kong et al., 2021; Fu et al., 2022; Xie et al., 2022).

2.5. ML and DL model justification

Employing a diverse array of 15 models—eight DL models for

vegetation cover prediction and seven more traditional ML models for high air pollution zone forecasting—provides substantial benefits for our research. This variety enhances the robustness of our findings, as convergence of results across models with different operational methodologies bolsters our confidence in their validity. It ensures a well-rounded interpretation of the data, taking advantage of the unique

capabilities of different models to provide both depth and breadth in analysis. The use of multiple models mitigates the risk of individual model biases, leading to more balanced and generalizable outcomes. Ensemble learning, which combines predictions from various models, often results in superior performance by drawing on the strengths and minimizing the weaknesses of each model. This method also serves as a cross-validation mechanism, where discrepancies between models can flag potential anomalies or overlooked insights. In addition, the diversity of models allows for flexibility in analysis. DL models may excel in complex tasks like image segmentation, while ML models may perform better in regression or time-series analysis. This multi-model approach also provides a safety net; if one model underperforms or overfits, the others maintain the integrity of the insights. As these models assess and prioritize features, we can identify key attributes that are consistently significant, shedding light on the most impactful factors. Ultimately, using 15 distinct models to investigate the link between vegetation cover and air pollution ensures a thorough and detailed exploration of the data. In terms of our model training, we used different ratios of training data to validation data for each model type. For ML it was 70% of the data to train the model and 30% for validation. However, for DL it was 80% of the data used for training and 20% for model validation based on the higher data demands for DL models, meaning more of the data was used to train the model. In addition, for DL, we utilized a lighter architecture, such as Resnet 18, and generated hundreds of chips from the images for training. We also employed a nine-variable ML approach to ensure that our data was sufficient to train the ML models.

3. Results

3.1. Analysis using machine learning algorithms

In this study, we implemented an array of machine learning

algorithms to predict areas of air pollution susceptibility across three classes of low, medium, and high likelihood (Fig. 6). In association with the spatial model results, we calculated Area Under the Curve (AUC) and Receiver Operating Characteristic (ROC) scores (Fig. 7). From these results we can evaluate the model performance and see Decision Tree (DT) and Naïve Bayesian (NB) lagged in performance compared to their counterparts. On the other hand, Extreme Gradient Boosting (XGB), Support Vector Machine (SVM), and Random Forest (RF) exhibited commendable performance. The ROC curve, a pivotal evaluation metric for classification models, graphically showcases the trade-off between sensitivity (true positive rate) and specificity (false positive rate) across varying classification thresholds. This visualization provides a holistic perspective on the model's performance, highlighting the equilibrium between sensitivity and specificity – a crucial factor that influences the selection of an optimal threshold tailored to distinct applications (Fig. 7).

Integral to the ROC curve is the AUC metric, which offers an unbiased measure of a model's overarching performance, irrespective of the chosen threshold. The AUC, essentially, quantifies the likelihood that the model ranks a randomly selected positive instance higher than a randomly chosen negative one. With AUC values spanning from 0 to 1, a score of 1 denotes perfect classification, 0.5 suggests random guessing, and 0 implies entirely erroneous predictions. Fig. 6 delineates the AUC-ROC values, reflecting the proficiency of various machine learning algorithms in aptly categorizing distinct air pollution intensities (Low, Medium, High) derived from remote sensing data. The SVM, XGB, RF, and NN algorithms demonstrate stellar performance, boasting high AUC-ROC scores (0.98–0.99) across all pollution categories. In contrast, while KNN showcases respectable performance, its AUC-ROC scores (0.96–0.98) lag slightly behind the leading quartet. The NB and DT algorithms anchor the lower end of the performance spectrum. Particularly, the NB algorithm, though exhibiting decent performance at heightened pollution levels (0.94), faces challenges at the medium level

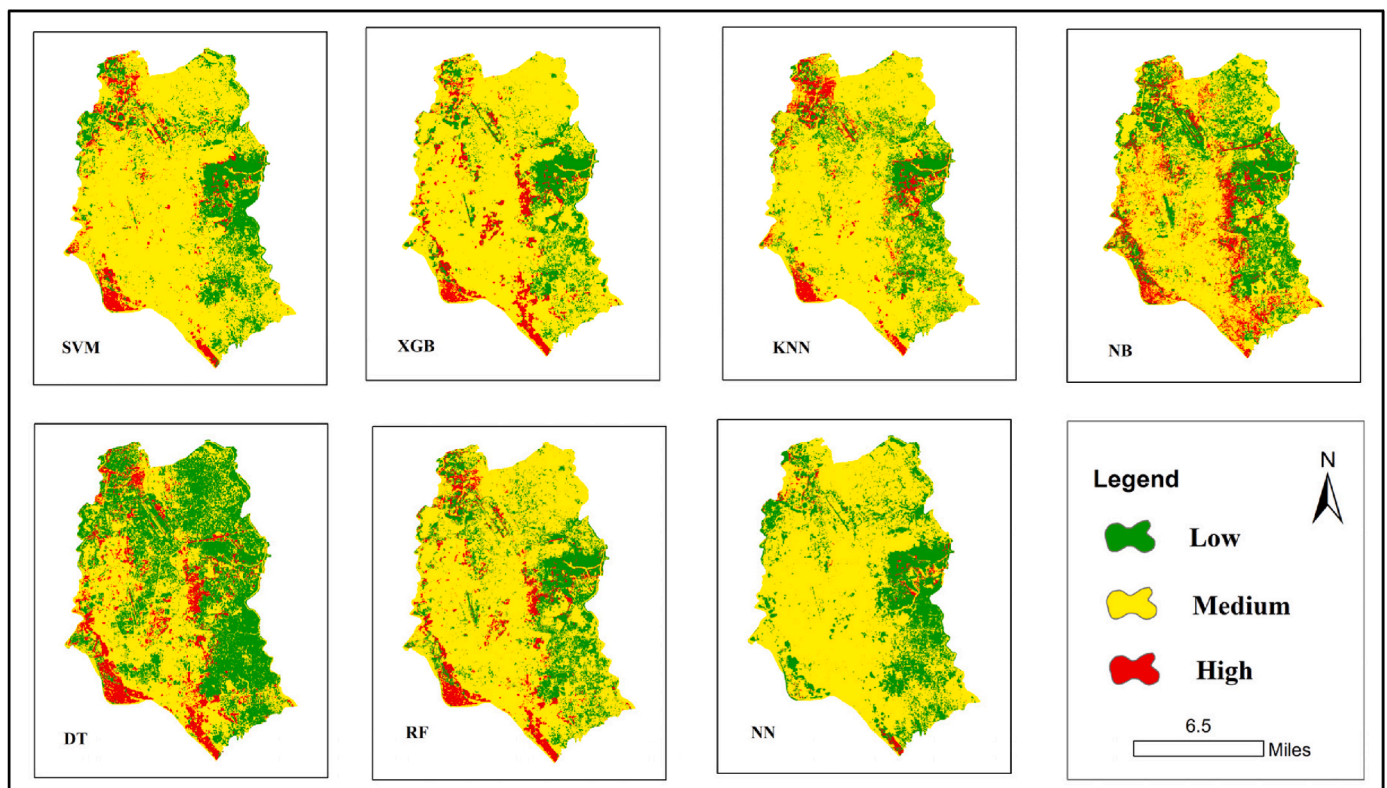


Fig. 6. Outcome of the 7 Machine Learning Models for the Prediction of the Air Pollution Susceptibility Zone. Models used are Support Vector Machine (SVM), Extreme Gradient Boosting (XGB), K- Nearest Neighbor (KNN), Naïve Bayesian (NB), Decision Tree (DT), Random Forest (RF), Neural Network (NN).

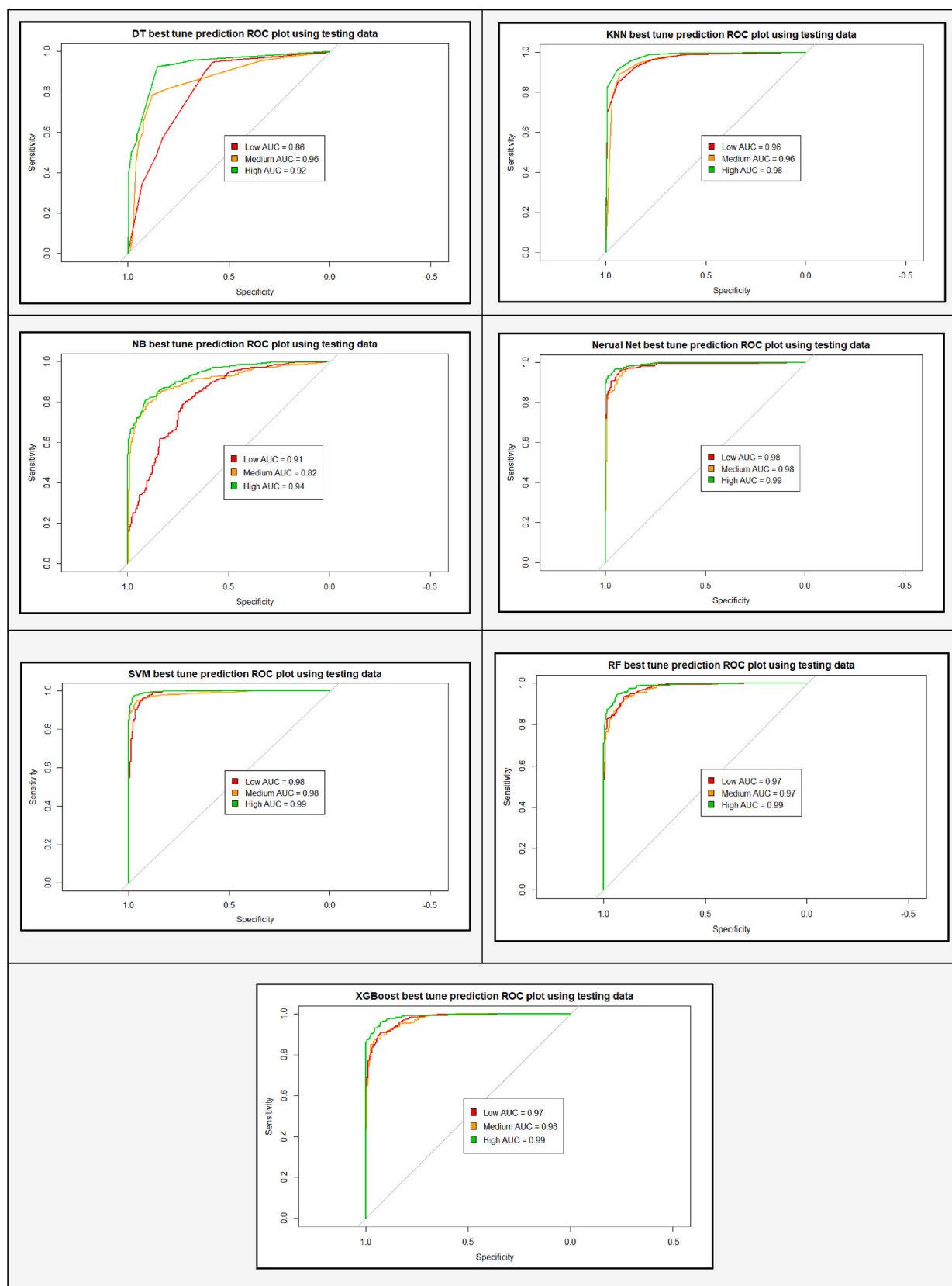


Fig. 7. Area Under the Curve (AUC) Receiver Operating Characteristic (ROC) curves for evaluating the performance of the 7 ML classification models predictions of high air pollution areas. Models used are Support Vector Machine (SVM), Extreme Gradient Boosting (XGB), K- Nearest Neighbor (KNN), Naïve Bayesian (NB), Decision Tree (DT), Random Forest (RF), Neural Network (NN).

(0.82). Similarly, the DT algorithm, though adept at medium pollution categorization (0.96), struggles at the extremities – both low (0.86) and high (0.92) pollution tiers. Considering these AUC-ROC scores, we can categorize the algorithms into three performance bands: Top-tier (SVM, XGB, RF, NN), Mid-tier (KNN), and Lower-tier (NB, DT). This stratification offers a transparent assessment of each algorithm’s capability in

monitoring air pollution through remote sensing data. Nevertheless, it is imperative to acknowledge that choosing an algorithm should also factor in additional considerations like computational demands, model interpretability, and the specific nuances of the application in question, ensuring a comprehensive and pragmatic application of these algorithms in air pollution surveillance.

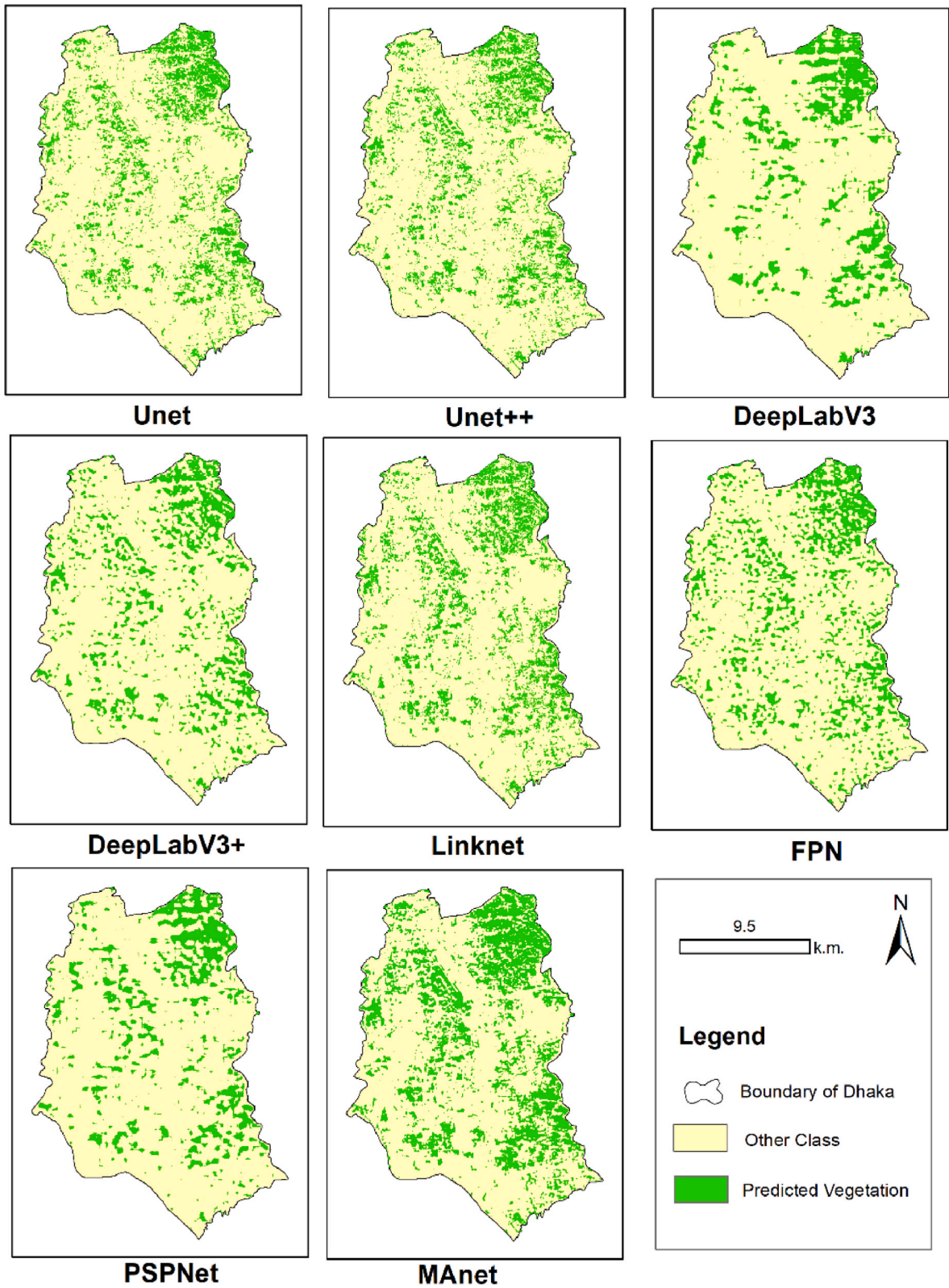


Fig. 8. Predicted vegetation from the deep learning models.

The superior performance of the ML models SVM, XGB, RF, and NN in our air pollution study reflects their capacity to manage complex, nonlinear relationships within environmental data. XGB, employing a gradient boosting framework, excels in refining predictions iteratively, which is beneficial when dealing with the complex interaction's characteristic of air pollution data. SVM's proficiency in projecting data into higher-dimensional spaces through the kernel trick enables effective handling of non-linear classification issues pertinent to air quality indices. RF leverages the collective output of numerous decision trees, enhancing predictive accuracy and robustness against overfitting - a key advantage when identifying diverse patterns and anomalies in air pollution trends. NN, with their deep learning capabilities, capture hierarchical feature representations crucial for interpreting both the detailed and overarching environmental factors affecting air quality. The astute selection of hyperparameters and training methods, along with these models' abilities to address data imbalances, likely contributed to their efficacy in analyzing the relationship between vegetation cover and air pollution.

3.2. Deep learning models to predict areas of high vegetation cover in the Dhaka region

The deep learning models utilized in the study yielded intriguing results in terms of their predictions of areas of high vegetation cover, as presented across Fig. 8, Tables 2 and 3. These results, when considered holistically, offer insights into both the models' training prowess and their ability to generalize on novel datasets. Fig. 8 displays the predicted vegetation cover for the Dhaka region as generated by different deep learning models. Each panel represents the output from one of the models: Unet, Unet++, DeepLabV3, DeepLabV3+, Linknet, FPN, PSPNet, and MA-net. Upon initial observation, there is a significant consistency across the models in terms of the broad patterns of vegetation distribution. Most models seem to highlight a scattered clustering of dense vegetation across the city. However, subtle differences can be noticed in the spatial extent and concentration of predicted vegetation regions among the models. For instance, models like Unet++ and DeepLabV3 appear to predict slightly denser vegetation clusters compared to, say, Linknet or FPN. Fig. 9 visualizes the predicted decreases in vegetation cover in the Dhaka region from 1990 to 2022, as deduced by various DL models. A cursory glance reveals that while there is a consensus across the models regarding certain areas experiencing a vegetation decline, there are also nuanced variations in the predictions

Table 2
Hyperparameters of machine learning and deep learning models.

Models	Hyperparameters
Deep Learning (General Structure)	ENCODER = "resnet18" ENCODER_WEIGHTS = 'imagenet' CLASSES = ["Green Space"] ACTIVATION = 'sigmoid' DEVICE = 'cuda' Epoch = 150, chip_size = 32, stride_x = 8, stride_y = 8, crop = 12, n_channels = 3 hidden = 3, threshold = 0.05
Neural Network	tuneLength = 10, preProcess = ['center', 'scale'], trControl = control, metric = 'Kappa'
Decision Tree	trControl = control
KNN	fl = 0, usekernel = True, adjust = 0.5, preProc = ['center', 'scale'], trControl = myControl, metric = 'Accuracy'
Naive Bayes	tuneLength = 10, preProcess = ['center', 'scale'], trControl = control, classProbs = True
SVM	nrounds = 200, max_depth = 18, eta = 0.03, gamma = 0.1, colsample_bytree = 1, min_child_weight = 0, subsample = 0.5, preProc = ['center', 'scale'], trControl = myControl, metric = 'Accuracy', tuneLength = 10, importance = True
XGBoost	tuneGrid = {'mtry': range(1, 21)}, trControl = trControl_rf, metric = 'Accuracy', importance = True
Random Forest	

Table 3

Loss, F1 Score, Precision, Recall and Accuracy scores of DL Models.

Models	Loss	F1 Score	Precision	Recall	Accuracy
Unet	0.6474	0.6667	0.6373	0.689	0.8941
Unet++	0.6722	0.6524	0.6136	0.6997	0.8892
MA-net	0.6619	0.6559	0.6306	0.6888	0.8932
Linknet	0.7155	0.624	0.5841	0.6829	0.9021
FPN	0.7900	0.5822	0.5234	0.6829	0.8619
PSPNet	0.8466	0.5609	0.4896	0.6587	0.8481
DeepLabV3	0.8873	0.5055	0.5011	0.5146	0.8593
DeepLabV3+	0.8209	0.5258	0.5258	0.5879	0.8672

of each model. For instance, the models Unet++ and DeepLabV3 predict a more dispersed pattern of vegetation decrease, with sporadic red patches scattered throughout. In contrast, models like Linknet and FPN seem to predict more concentrated clusters of decline in specific regions. The MA-net model portrays a somewhat denser clustering of vegetation decrease compared to others. Overall, these visuals highlight the areas within Dhaka that have been most affected in terms of vegetation loss over the last three decades, as per the predictions of the DL models. The differences among the models offer a comprehensive view of potential zones of concern and provide a multi-faceted perspective on vegetation dynamics in the region.

Turning to the model evaluations in Table 2, the metric "Loss" stands out as an initial point of discussion. In essence, the loss is a measure of how well the model's predictions conform to the actual outcomes during the training phase. A lower loss suggests that the model has effectively captured the nuances and intricacies of the training data. In this context, Unet demonstrated the lowest loss, potentially signaling its proficiency during the training phase. Conversely, DeepLabV3 recorded the highest loss, indicating some potential challenges in modeling the training data. Moving forward, "Accuracy" in Table 3 measures the ratio of the model's correct predictions to the total predictions made. Linknet emerged as a leader with an impressive accuracy of 0.9021, only slightly edging out models like Unet and MA-net, which also showcased commendable performance. The F1 score, another key metric from Table 3, encapsulates the balance between precision (how many of the predicted positive instances are truly positive) and recall (how many of the actual positive instances were correctly predicted by the model). Here, Unet is the top performing model, reflecting its efficiency in achieving this equilibrium.

Bringing in Table 3, which introduces the models' accuracy on validation data from the 2021 image (previously unseen by the models), we gain insights into their generalization capabilities. Generalization is a model's ability to perform well on new, unseen data, and it is a cornerstone for any practical application. Unet, Unet++, and Linknet all demonstrated a stellar accuracy rate of 0.95, suggesting that they not only trained well but also generalized effectively to new data. Other models, including MA-net, FPN, PSPNet, and DeepLabV3, all hovered closely with an accuracy of 0.94. DeepLabV3+ trailed slightly with an accuracy of 0.93, though this is still a commendable score.

In analyzing the performance metrics, Unet and Linknet emerge as leading DL models, achieving both efficient training and strong generalization capabilities. The collective contributions of all models are significant, displaying proficiency in different aspects of the analysis. Model selection should be informed by various considerations, including the application's specific needs, computational resources, and interpretability. Unet and Linknet's effectiveness is likely due to their architectures, which are well-suited to processing spatial data, allowing for the detection of local and global vegetative patterns. Their multi-layered structure helps in discerning complex patterns, crucial for high-resolution image segmentation - a key element in this study. These models' performance is also enhanced by their ability to learn from large datasets, which could lead to better generalization of new data. Additionally, the training methodologies, including the optimization of hyperparameters and learning rates, may have been particularly well-

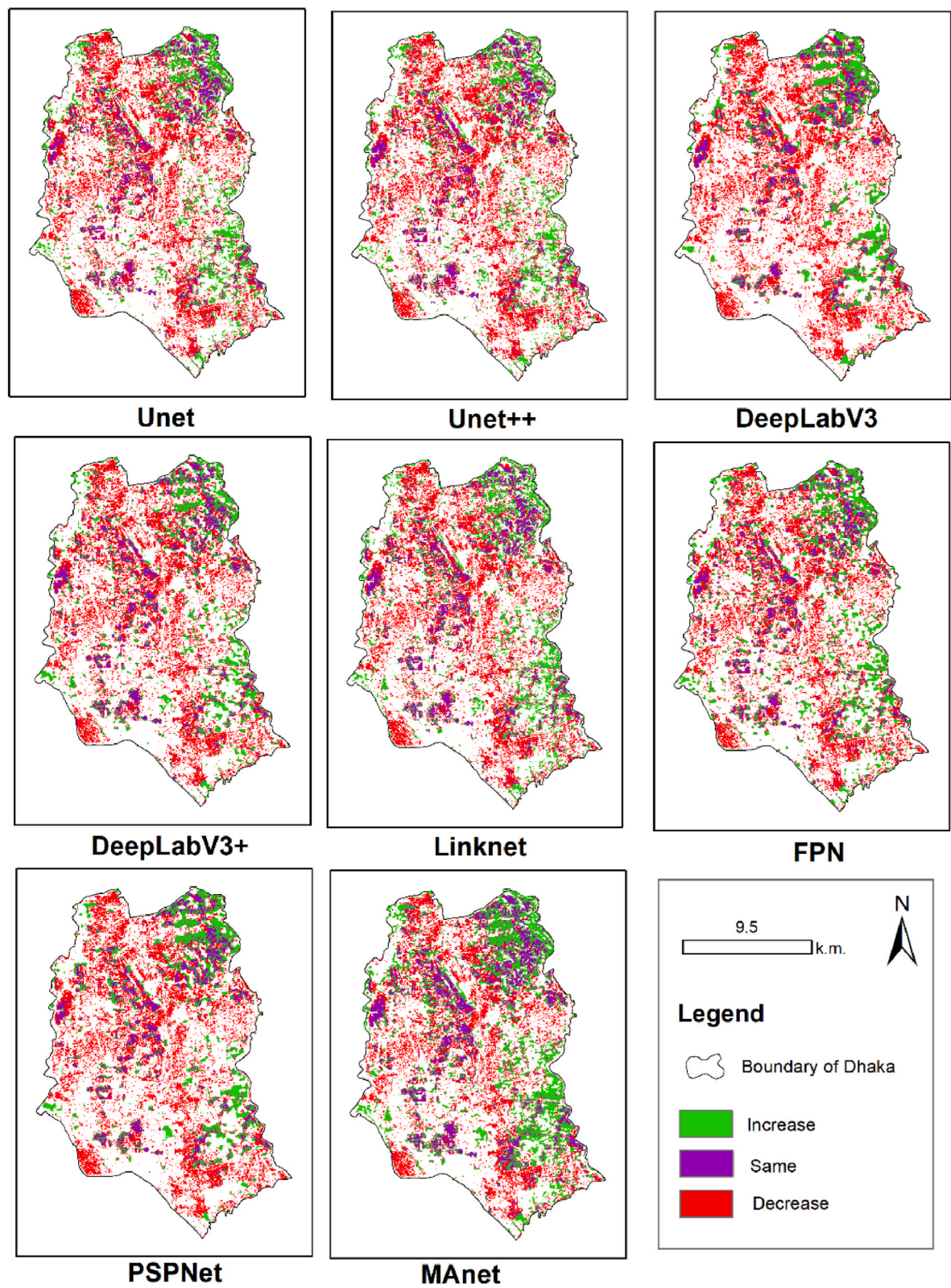


Fig. 9. Comparison of 1990 vegetation cover and 2022 predicted vegetation cover red highlights vegetation loss by 2022.

adjusted for these models, contributing to their standout performance in segmenting vegetation cover in Dhaka. Iterative refinement in DL, where models gradually improve through multiple iterations, could have further honed these models' capabilities, resulting in their distinguished performance in the study.

3.3. Relationship between ML predicted areas of pollution and DL predicted vegetation abundance

Finally, we can evaluate the spatial model predictions that showcase the predicted zones of air pollution from the ML models (Fig. 6) and the areas of vegetation decline from the DL models (Fig. 8). From this

comparison several patterns emerge that allow us to draw conclusions about the relationship between air pollution and vegetation loss in Dhaka from 1990 to 2022. First and foremost, in areas where the ML models predict high air pollution (as indicated by the red regions in Fig. 6), there is a marked overlap with the areas indicating a decline in vegetation in the DL models' predictions (Fig. 8). This overlap is especially evident in the outputs from models like SVM, XGB, and RF for air pollution, and Unet, Unet++, and DeepLabV3 for vegetation decline. Furthermore, it is noticeable that the regions with medium air pollution (shown in yellow) also display some correspondence with areas of vegetation decrease, though not as pronounced as the high air pollution zones. This suggests a possible gradient relationship, where areas with the highest air pollution experience the most significant vegetation decline, while regions with medium air pollution witness a moderate decrease in vegetation. It is also worth noting that areas predicted to have low air pollution (shown in green on Fig. 9) generally have less overlap with the predicted decrease in vegetation (Fig. 9). This may indicate that while air pollution is a significant factor affecting vegetation decline, other factors could also play a role in areas with lower air pollution levels. In conclusion, the visual examination of both images

(Figs. 6 and 9) offers compelling evidence that high air pollution zones indeed overlap significantly with areas of vegetation loss from 1990 to 2022. This relationship underscores the detrimental impacts of air pollution on urban green spaces and the ecosystem at large.

3.4. Relationship between the ML and DL models, or specifically the ML predicted high air pollution zones and the DL predicted vegetation areas for Dhaka

Fig. 10 shows distinct predictions from three ML models and three DL models visualized over the geographic expanse of Dhaka. The representation is straightforward: ML model output shows areas predicted to experience high air pollution, and are highlighted in red, whereas the DL model predictions, highlighting zones with predicted higher vegetation cover are depicted in green. The DL model predictions, specifically from U-Net, U-Net++, and MA-Net, highlight predictions of vegetation cover where these models anticipate notable vegetation cover. The congruence in predictions across these DL models suggests a high degree of agreement among them regarding the distribution of vegetation within the study area. Conversely, the ML models—RF, SVM, and

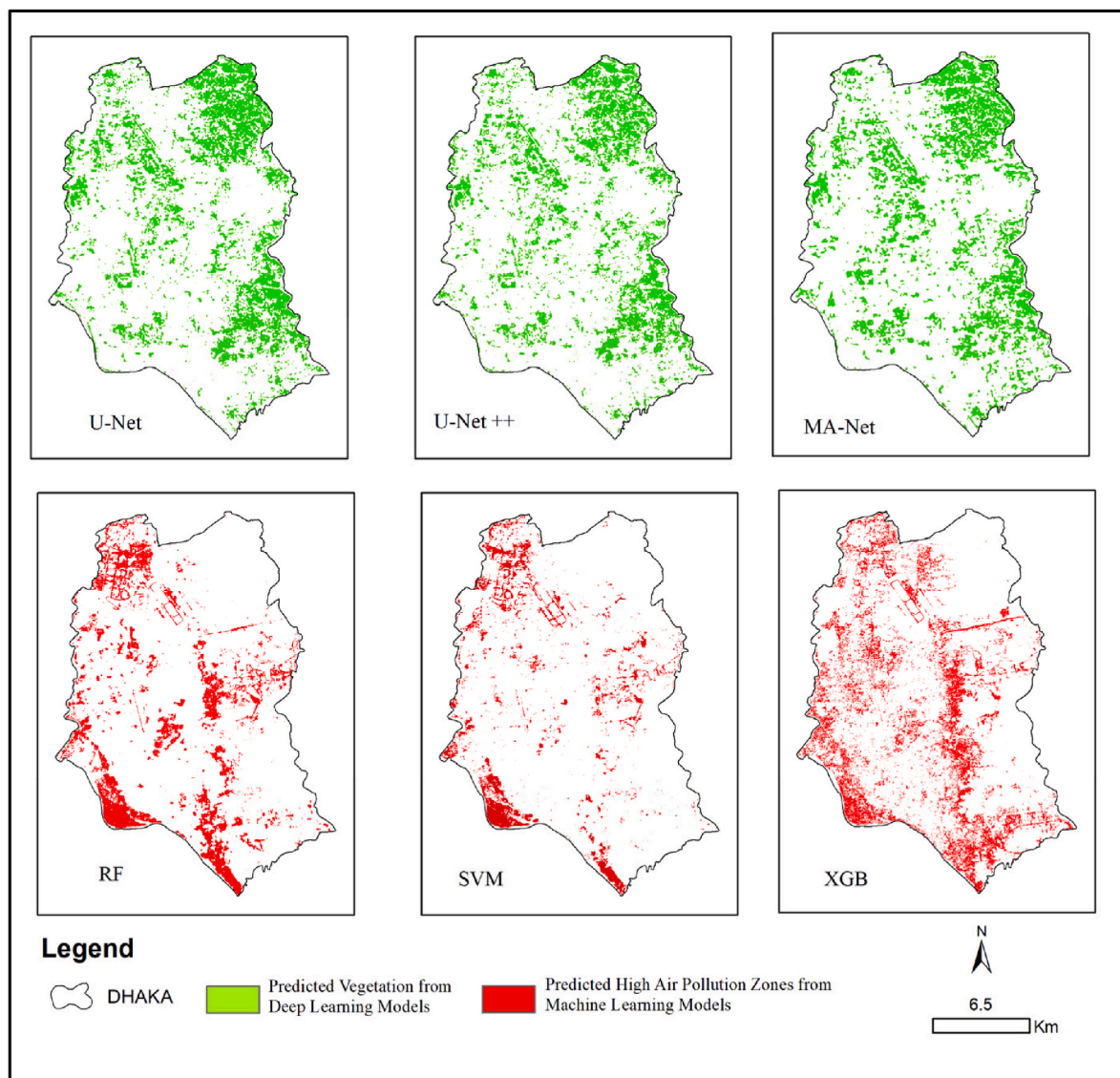


Fig. 10. Spatial predictions made by three Machine Learning (ML) models and three Deep Learning (DL) models for Dhaka, Bangladesh. The displayed maps are delineated into two distinct categories based on their predictions: ML model output of areas with predicted high air pollution (depicted in red) and DL model output of areas with predicted high vegetation cover (shown in green).

XGB—indicate predictions denoting regions where these models foresee elevated levels of air pollution. There is more spatial variability in the areas of high air pollution zones across the ML model predictions, when compared to the more consistent DL model predictions.

When examining the spatial relationship between the DL and ML model predictions, an inverse spatial relationship is evident. Regions with dense vegetation, as indicated by the DL models, tend to correspond to areas that the ML models do not flag as high air pollution zones. This observation suggests that areas rich in vegetation might play a mitigating role in controlling or reducing air pollution levels. On the other hand, regions sparse in vegetation seem more susceptible to heightened pollution levels. This relationship between vegetation and air pollution, as indicated by the model predictions, underscores the potential environmental interplay between green cover and air quality within Dhaka. Given the importance of these relationships further statistical analysis on the relationships between the ML high air pollution predictions and the DL high vegetation predictions is necessary.

The provided heatmap (Fig. 11) visualizes the correlation matrix for the three ML high air pollution predicted surfaces and the three DL high vegetation cover predicted surfaces (Fig. 10). This correlation is a representation of the linear relationship between the predictions of different model outputs. In terms of comparing the ML models with each other, this allows us to better determine the overlap in predictions across these three models in terms of predicting areas of high air pollution. Similarly, when comparing the three DL models with each other we can then evaluate the similarity of spatial prediction of areas of high vegetation cover. In addition to comparing the surfaces predicted across the same categories we can also relate the output of the ML models to those of the DL models and thus provide a statistical discussion (Fig. 10) to relate to the spatial discussion (Fig. 9), in terms of how areas of high predicted air pollution relate spatially to areas of high vegetation cover.

The ML models were employed to predict regions within Dhaka with anticipated high air pollution levels. Overall, from Fig. 10 we can see that the SVM model predicts the lowest extent of area as high air pollution zones and XGB predicts the greatest extent of high air pollution zones. Areas of high air pollution from SVM are also predicted by the other two ML models, but they extend the areas of prediction. When looking across the pairs of models we can see that for SVM and RF, they display a moderately positive correlation of 0.51, suggesting that their predictions of high air pollution areas often align. For SVM and XGB

there is a correlation of 0.23 between these models which implies a weaker agreement in their air pollution predictions. RF and XGB have a correlation of 0.42 which denotes that these two models moderately agree in identifying areas of high air pollution.

The DL models, namely MAnet, U-Net, and U-Net++, were utilized to predict areas of vegetation cover within the Dhaka study region. The MAnet model, when predicting the location and extent of vegetation cover, has a strong agreement with both U-Net (0.63) and U-Net++ (0.61). U-Net and U-Net++ correlation indicates their vegetation predictions are highly aligned, as denoted by their strong correlation of 0.8. Within their respective categories, however, the models display varying degrees of agreement, with U-Net and U-Net++ showcasing particularly strong concordance in predicting vegetation areas.

When comparing the results of DL models with ML models, we may examine the correlation between the anticipated air pollution zones (output of the ML model) and the predicted decreased vegetation cover (output of the DL model). The DL models generally showcase negative correlations, ranging between -0.1 and -0.21 , with the ML models. Spearman's rank correlation coefficient scales from -1 to 1 . A perfect negative correlation is -1 , no correlation is 0 , and a perfect positive correlation is 1 . The number between -0.1 and -0.21 , indicates a weak to moderate negative association, which although not the strongest numerically, makes sense biologically, with higher air pollution regions resulting in lower vegetation cover. This suggests that areas predicted to have high air pollution by the ML models are inversely related to the areas predicted to have high vegetation cover by the DL models. In essence, the ML models' predictions about areas of high air pollution in Dhaka often seem to contrast with the DL models' predictions about vegetation cover. The overarching interpretation suggests that areas with high air pollution, as identified by the ML models, tend to have less vegetation, as indicated by the DL models. This could suggest that areas with dense vegetation might play a role in reducing or mitigating air pollution levels, acting as natural air purifiers. Conversely, areas with scant vegetation might lack this natural barrier and hence exhibit elevated levels of pollutants.

3.5. Model evaluation and study limitations

Within the framework of this research, we analyzed a series of ML and DL models and their output. As such, validation of results in terms of consensus of patterns and outputs allows for us to understand and validate our study (Figs. 6 and 8). Within ML and DL models, sensitivity analysis is not common, in part due to the significant input demands of ML and specifically, DL models (Southworth et al., 2024). For our research ML models were analyzed in terms of their Area Under the Curve (AUC) Receiver Operating Characteristic (ROC) curves for evaluating the performance of the 7 ML classification models predictions (Fig. 7), whereas for the DL models Loss, F1 Score, Precision, Recall and Accuracy were calculated and compared (Table 3). In addition, after model evaluation the DL model was then run on a completely independent dataset, 2021, which was not part of the initial study, in order to validate model performance (Table 4; Southworth et al., 2024). The establishment of rigorous techniques and assessment frameworks is

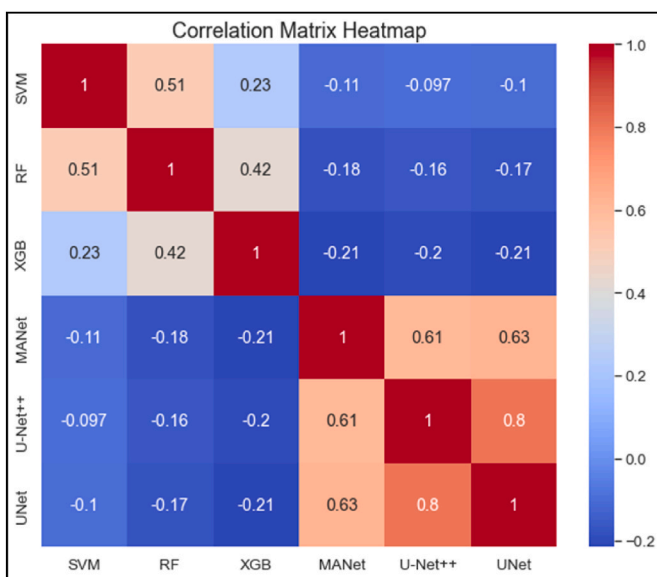


Fig. 11. Spearman Correlation Matrix of the best Machine Learning models for Air Pollution Prediction compared to the top Deep Learning Models for vegetation cover prediction to indicate their relationship over the study area.

Table 4

Validation of Deep Learning models using images from the year 2021.

DL Models	Accuracy
Unet	0.95
Unet++	0.95
MAnet	0.94
Linknet	0.95
FPN	0.94
PSPNet	0.94
DeepLabV3	0.94
DeepLabV3+	0.93

crucial to maintain the relevance and efficacy of models in practical applications, hence facilitating improved insights and decisions derived from remote sensing data (Adegun et al., 2023). The current research in this domain is to develop frameworks that evaluate model robustness and enhance the clarity of communication on assumptions and implications derived from such studies.

The validity and reliability of the findings reached in the study may be impacted by several limitations when comparing model results to observational data. First, fundamental variations in the geographical and temporal resolutions of the datasets employed may cause disparities between model predictions and actual data. Satellite-derived data frequently differs in scale and could not match ground-based observations exactly, which could result in inaccurate depictions of the real situation. The evaluation procedure may also be made more difficult by measurement mistakes, biases, and inconsistencies in the observational data itself. Overfitting, which may happen when models are too customized to the training data and lead to poor generalization to new or unexplored datasets, is another drawback of predictive models. The uncertainty surrounding model predictions and observational data may not be sufficiently addressed in the study. Again, within this research, our use of multiple ML and DL models and their comparisons and consensus of results, along with our rigorous evaluations and comparative statistical analysis of performance, allows us to have confidence in our findings and the relationships proposed across our models (Figs. 10 and 11). Finally, the ability of our DL model to be applied to the 2021 dataset and perform well, highlights that our DL models were able to predict accurately from unknown data and therefore were not constrained by overfitting (Table 4). While all analyses based on remotely sensed data are limited by the input data used and validation data available, the model consensus across our multiple ML and DL models, ability to predict 2021 data and strength of statistical analysis of model performance, all indicate a study robustness and clarity of findings.

4. Discussion

Dhaka, the capital of Bangladesh, has undergone rapid growth, boosted its economy but also caused serious environmental problems. This growth, important for economic development, is harming the local environment. Air pollution in Dhaka has been a problem for a long time, getting worse in recent years due to increased industrial activities and more vehicles. This increase in urban development is reducing the city's green areas. The expansion of buildings and roads are replacing the natural vegetation. Research by Byomkesh et al. (2012) and Abdullah et al. (2022) points out that this pattern of development is leading to various environmental issues. To plan for sustainable cities, it is important to predict future pollution levels, especially as past trends show increasing emissions.

The study provides an accurate prediction of the effects of PM_{2.5} and PM₁₀ on urban vegetation. Different vegetation types display unique reactions to air pollution (Han et al., 2022; Jim and Chen, 2008). For example, deciduous trees like oaks and maples might exhibit observable harm to their foliage, decreased photosynthetic efficiency, and restricted development as a result of their extensive leaf surface and extended lifespan, rendering them more vulnerable to pollution. Coniferous trees, such as pines and spruces, can experience different physiological pressures, which can cause them to shed their needle-like leaves and slow down their growth. In addition, herbaceous plants and grasses can show different levels of sensitivity to pollutants, with some species being more tolerant while others undergo significant growth inhibition and chlorosis.

Research has shown that vegetation and green spaces within urban landscapes, can act to reduce air pollution concentrations, specifically of PM (Diener and Mudu, 2021). Plant communities that exhibit a high degree of stratification and possess distinct leaf types are especially effective in reaching this outcome (Diener and Mudu, 2021; Zhu et al., 2019). In addition, the study discovered strong positive connections

between the magnetic characteristics of tree leaves and the levels of metals in PM_{2.5}. This implies that analyzing leaf magnetism could be a feasible approach for assessing the presence of heavy metal pollution in the air, if suitable models are employed (Leng et al., 2018). Research from the treeless street canyons of central Shanghai, illustrates the concentration of PM_{2.5} particles decreasing as the height increases. However, the presence of trees obstructs this dispersion process. Canopy density, leaf area index (LAI), wind speed, pruning intensity, and tree species are all factors that influence the decrease of PM_{2.5}. Research by Jin et al. (2014) suggests that an optimal canopy density of 50%–60% and LAI of 1.5–2.0 should be considered in urban planning for maximum effectiveness. Gaining insight into these diverse impacts provides a more comprehensive understanding of how air pollution affects urban green areas and helps identify resilient species that can withstand high pollution levels. Gaining a more nuanced understanding of these mechanisms is crucial for urban planning and the development of programs to enhance the resilience of urban vegetation to such pollution challenges (Diener and Mudu, 2021; Jim and Chen, 2008; Lu et al., 2024). Acquiring this knowledge will ultimately result in urban settings that are both healthier and more sustainable, although further field studies are still needed to completely understand the role of vegetation in impacting air pollution in urban landscapes (Han et al., 2022; Su et al., 2022). This will ensure that urban vegetation can efficiently reduce air pollution while flourishing in contaminated areas.

The analysis of ML and DL models offers a comprehensive view of Dhaka's environmental trajectory, particularly focusing on high air pollution zones and vegetation cover. The ML models, specifically the SVM, RF, and XGB, effectively predicted areas marked by heightened air pollution within the city. Among them, SVM and RF displayed particularly strong performances, suggesting their utility in modeling complex urban pollution dynamics. Conversely, the DL models, especially Unet, Linknet, and Unet++, emerged as frontrunners in accurately predicting vegetation cover across Dhaka. These models not only showcased prowess in capturing training data nuances but also demonstrated impressive generalization capabilities when tested on unseen 2021 data. Synthesizing the outputs of both model types revealed a concerning trend: areas with diminishing vegetation often coincide with zones exhibiting increased air pollution, underscoring a potential negative feedback loop. This synthesis, drawing from both ML and DL outputs, paints a comprehensive picture of Dhaka's environmental challenges, emphasizing the urgent need for sustainable interventions.

Upon closely analyzing the predictive outputs from both the machine learning models (highlighted in Fig. 5) and the deep learning models (as depicted in Fig. 7), a striking correlation becomes evident. The zones characterized by high air pollution predominantly coincide with regions that have experienced a marked decline in vegetation from 1990 to 2022. This spatial overlap accentuates the profound adverse effects that escalating air pollution levels can exert on urban green spaces. Such findings underscore the urgent need for integrated urban planning and pollution control measures to safeguard and potentially rejuvenate the diminishing urban vegetative cover. The ML and DL models' outputs unanimously corroborate a concerning prognosis for Dhaka's green spaces. They signal a persistent decline in vegetation, concomitant with surging levels of PM_{2.5} and PM₁₀. Such patterns resonate with the city's developmental narrative, where industrial growth and infrastructural expansion have often overshadowed environmental considerations. The urban sprawl, an upshot of burgeoning population density, has invariably led to forested regions succumbing to infrastructural demands. But beyond this evident cause-effect relationship between urbanization and deforestation lies a more covert, yet potent, agent of change – air pollution. Chronic exposure to pollutants has debilitating effects on plant health, manifesting in diminished biomass and vigor. Furthermore, the ebbing green cover, which traditionally acted as Dhaka's ecological shield, intercepting and absorbing pollutants, now seems to be in a retreating mode, further exacerbating air quality concerns.

Investigating the linkage between air pollution and vegetation presence can be done using ML and DL methodologies. These methodologies can help quantify the changes in air pollution and assess its impact on vegetation. ML methods, such as dimensionality reduction, can extract important information from air pollution data and summarize it to understand the complex environment (Kuo et al., 2022). DL algorithms, like LSTM models, can be used to assess the concentrations of pollutants in the atmosphere and analyze their effects on vegetation. By leveraging these techniques, policymakers and researchers can better understand the relationship between air pollution and vegetation and make informed decisions to mitigate its impact on the environment and human health. Rising air pollution, specifically PM_{2.5} and PM₁₀, is linked to decreasing vegetation cover. Studies have shown that tree coverage in urban areas can significantly reduce PM₁₀ and PM_{2.5} levels, leading to improved air quality (Sierra-Porta et al., 2023). Increasing vegetation coverage has been found to have a significant impact on reducing PM_{2.5} levels (Rahmani et al., 2020). Different land-cover types and patterns also affect PM₁₀ and PM_{2.5} concentrations, with bare soil having the highest PM₁₀ levels and high canopy density vegetation having the highest PM_{2.5} levels (Fan et al., 2019). The design of land cover structure plays a crucial role in ameliorating air particle concentrations in the microscale environment (Agathokleous et al., 2020). Overall, these findings suggest that increasing vegetation cover can help mitigate the negative effects of air pollution, particularly PM_{2.5} and PM₁₀ (Russell and Dyer, 2017).

ML and DL methods can be used to understand the relationship between air pollution, specifically PM_{2.5} and PM₁₀, and decreasing vegetation cover. The use of these methods allows for the quantification of changes in air pollution and the identification of the most suitable estimation methods for policymakers and directors. Additionally, these techniques can aid in the accurate prediction of long-term PM_{2.5} concentrations, which is crucial for reducing air pollution and avoiding its negative consequences (Alzwy et al., 2023). Furthermore, DL models, such as the graph convolutional network (GCN) model, can be employed to predict the level of air pollution in smart environments, contributing to the planning and design of smart cities (Ayus et al., 2023). By combining ML and DL algorithms, the relationship between air pollution and decreasing vegetation cover can be better understood, leading to more effective strategies for mitigating the impact of air pollution on the environment (Singh et al., 2023).

Although machine learning (ML) and deep learning (DL) models have impressive predictive skills, it is crucial to thoroughly analyze and understand the interpretation of these model outputs to improve their practical use. Gaining comprehension of how AI-generated forecasts might be transformed into practical observations is essential for urban planning and environmental management. Historically driven by government initiatives, environmental governance can significantly benefit from AI's data analysis capabilities, facilitating multi-stakeholder approaches and minimizing biases in decision-making (Nishant et al., 2020). The emergence of AI in urban planning highlights the need for interdisciplinary approaches to integrate smart and eco-urbanism, addressing empirical sustainability difficulties and environmental concerns (Palmini and Cugurullo, 2024). Sophisticated air quality monitoring systems in smart cities, powered by machine learning, ensure timely responses to air quality deviations, enhancing urban living and sustainability (Jha et al., 2021). Furthermore, transparent carbon footprint labeling, and stricter regulations are essential to mitigate the ecological damage from AI technologies' high energy consumption (Brevini, 2020). By synthesizing these perspectives, we can develop more effective strategies for promoting sustainable urban development.

The City Brain initiative exemplifies the practical implementation of AI in local government administration, leading to substantial progress in the digitalization and automation of processes (Bibri et al., 2024b). The combination of AI and artificial intelligence of things (AIoT) is crucial in addressing environmental degradation and climatic disruptions in the context of "smarter eco-cities." This emphasizes the significance of AI in

promoting sustainable urban development (Bibri et al., 2024a). The ability of ML and DL to scale and be applied in various contexts allows AI to overcome the constraints and expenses associated with early knowledge-based systems. This, in turn, facilitates logical decision-making based on data (Bickley et al., 2024). In order to effectively utilize AI and accomplish Sustainable Development Goals (SDGs), enterprises need to develop supportive infrastructures, implement temporal sustainability indicators, and foster eco-centric innovation (Pan and Nishant, 2023). By placing emphasis on the interpretation of AI model outputs, it guarantees that predictive insights are efficiently employed in urban and environmental governance initiatives.

To significantly enhance the study's broader applicability and relevance, it may potentially be beneficial to include comparative assessments with other cities facing similar environmental challenges, such as Delhi, Beijing, and Mexico City. Metropolises like Dhaka encounter formidable obstacles because of heightened air pollution levels and rapid urbanization. These characteristics contribute to an optimal setting for assessing and contrasting the efficacy and outcomes of these models. Performing comparative evaluations will not only showcase the robustness and adaptability of the model in different urban contexts but also yield a comprehensive understanding of urban environmental deterioration. This methodology would streamline the process of identifying widespread patterns and urban-specific factors, ultimately resulting in potentially more effective and universally applicable measures for regulating air quality and vegetation deterioration. Furthermore, by testing the models using independent datasets or other urban environments, we may evaluate their generalizability outside Dhaka. This will improve their usefulness in worldwide environmental study and would reinforce the view that this study is reproducible, and the algorithms useful to different urban regions (Zhang et al., 2024).

Green spaces significantly reduce airborne PM levels, and this effect changes depending on the size, location, and type of vegetation. There is a lack of connection between relevant fields, so we need a research framework that provides better health guidance (Lu et al., 2024). An integrated approach focusing on PM reduction through green spaces is urgently needed (Li et al., 2023; J. Zhang et al., 2024). To tackle these complex issues, we need a comprehensive strategy that includes understanding current trends and making accurate forecasts. Monitoring how vegetation changes and predicting air pollution levels are key to creating effective policies for Dhaka's sustainable future. This study shows the importance of considering both spatial and temporal aspects of air quality and vegetation. Future research should expand this framework, using wider geographical data and improving prediction models, not just for Dhaka but for other fast-growing cities worldwide.

5. Conclusion

In this comprehensive study, ML and DL methodologies were employed to understand the relationship between air pollution and vegetation cover, and the deterioration of urban vegetation in Dhaka, Bangladesh. The findings underscore the detrimental impacts of increased particulate matter on urban green spaces. In predicting the potential effects of PM_{2.5} and PM₁₀ on urban vegetation, the XGB, SVM, and RF models emerged as superior, as evidenced by their enhanced accuracy and AUC-ROC metrics. Conversely, in the realm of DL, the Unet, Unet++, MAnet, and Linknet models achieved high F1 scores, showcasing their efficacy in anticipating alterations in urban vegetation. This research's novelty and conclusions comes from the combination of sophisticated DL and ML methodologies to examine the intricate interactions between air pollution and vegetation utilizing remotely sensed data. This work offers distinctive contributions to environmental science and remote sensing by integrating advanced approaches with a thorough approach to data representation. This research primarily introduces the utilization of DL segmentation models, including Unet, Unet++, MAnet, and Linknet, to improve feature extraction and classification of vegetation health indicators, such as

NDVI. This method facilitates a more refined comprehension of the effects of differing concentrations of air pollutants, particularly fine particulate matter (PM_{2.5}), on plant physiological responses throughout time. In contrast to standard models, DL enables the examination of complex patterns in extensive datasets, uncovering insights that may have previously been overlooked owing to the constraints of conventional methods.

This study highlights the serious effects of increased air pollution, both in the immediate area and nearby regions. It shows the negative environmental impact of uncontrolled city growth and resulting air pollution. Given the role of vegetation in potentially minimizing the impact of such PM pollution, the importance of increasing green spaces and better planning and integration of vegetation within urban landscapes, becomes of paramount importance. This study concludes that including pollution control via vegetative cover and plantings, in urban green space planning, is critical. Future research using high-resolution images could give more detailed information on the spatial and structural impacts of vegetation types and morphologies, to improve such planning potential even more. In addition, combining social and economic data with ML and GIS could deepen our understanding of how air pollution, urban growth, and green spaces are connected. This research contributes to the global conversation about environmental changes. It stresses the urgent need to reduce pollution and the importance of focusing on green spaces in city planning. The findings underline the importance of using spatially explicit data to guide urban development to produce urban landscapes which can minimize air pollution impacts for urban populations, and thus reduce the associated environmental pollution risks associated with such locations globally.

CRediT authorship contribution statement

Mashoukur Rahaman: Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Formal analysis, Conceptualization. **Jane Southworth:** Writing – review & editing, Writing – original draft, Supervision. **Amobichukwu Chukwudi Amanambu:** Writing – review & editing. **Bewuket B. Tefera:** Writing – review & editing, Writing – original draft. **Ali R. Alruzuq:** Writing – review & editing. **Mohammad Safaei:** Writing – review & editing, Writing – original draft. **Md Mueyed Hasan:** Writing – review & editing. **Audrey Culver Smith:** Writing – review & editing.

Ethical standards

The authors declare that the research article is compliance with the ethical standards of this journal.

Informed consent

The article does not contain any information that requires informed consent.

Availability of data and material

All data supporting the content of this manuscript are sourced in the reference list.

Declaration of AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to improve readability and language. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Data availability

Data will be made available on request.

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